KEY SHEET

MATHEMATICS

1	**A**	2	**B**	3	**A**	4	**A**	5	**ACD**	6	**BC**
7	**BC**	8	**6**	9	**7**	10	**1**	11	**36**	12	**2**
13	**7**	14	**2**	15	**0**	16	**6**	17	**6.75**		

PHYSICS

18	**A**	19	**B**	20	**A**	21	**A**	22	**ABC**	23	**BD**
24	**ABC**	25	**266**	26	**17**	27	**0**	28	**1**	29	**4**
30	**2**	31	**2**	32	**0**	33	**0**	34	**1**		

CHEMISTRY

35	**A**	36	**C**	37	**B**	38	**A**	39	**ABCD**	40	**ABC**
41	**AC**	42	**7**	43	**8**	44	**5**	45	**5**	46	**3**
47	**7**	48	**2**	49	**4**	50	**1.5**	51	**5135**		

SOLUTIONS
MATHEMATICS

1. $\frac{1}{n}\sum_{i=1}^{n}(x_i-2)^2=10$ And $\frac{1}{n}\sum_{i=1}^{n}(x_i+2)^2=18$

$\Rightarrow \sum_{i=1}^{n}(x_i+2)^2+\sum_{i=1}^{n}(x_i+2)^2=28n$ and $\Rightarrow \sum_{i=1}^{n}(x_i+2)^2-\sum_{i=1}^{n}(x_i-2)^2=28n$

$\Rightarrow 2\sum_{i=1}^{n}x_1^2+8n=28n$ and $\Rightarrow 8\sum_{i=1}^{n}x_I+8n\Rightarrow \sum_{i=1}^{n}x_i^2=10n$ and $\sum_{i=1}^{n}x_n=n$

2. We have $|f(x)-f(y)|\le|x-y|^{100}$

$\Rightarrow \left|\frac{f(x)-f(y)}{x-y}\right|\le|x-y|^{99}$

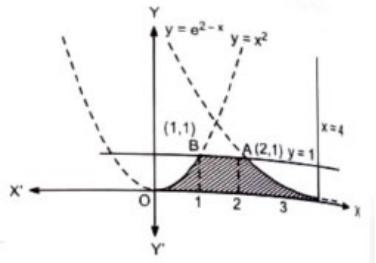
Now, $\Rightarrow \left|\lim_{y\to x}\frac{f(x)-f(y)}{x-y}\right|\le\lim_{y\to x}|x-y|^{99}$

$\Rightarrow |f'(x)|\le 0$

i.e $f'(x)=0$

$\Rightarrow f(x)=c$ (constant)

Given, $f(1000)=1\Rightarrow c=1\therefore f(x=1)$



Now, $\min.\{f(x),x^2,e^{2-x}\}=\min.\{1,x^2,e^{2-x}\}$

$\therefore$ The required area $=\int_0^1 x^2dx+(2-x)\times 1+\int_2^4 e^{2-x}dx$

$=\left[\frac{x^3}{3}\right]_0^1+1+e^2\left[\frac{e^{-x}}{-1}\right]_2^4=\frac{1}{3}+1+e^2\left(-e^{-4}+e^{-2}\right)$

$=\frac{4}{3}+1-e^{-2}=\left(\frac{7}{3}-\frac{1}{e^2}\right)$sq.units

3. $f(x)=2x+[x]+\sin x.\cos x$

4. Partial differentiate with respect to y Then put $x=0$. $f'(y)=\frac{1}{1-y^4}$

5. $I=-\frac{1}{8}\int\frac{t-3}{t^3}dt=-\frac{1}{8}\int\frac{1}{t^2}-\frac{3}{t^3}$

$=-\frac{1}{8}\left(\frac{1}{t}-\frac{3}{2t^2}\right)+e=\frac{2t-3}{16t^2}=\frac{8\cos x+3}{16(3+4\cos x)^2}+e$

$\Rightarrow a=3,b=8,c=16,d=2$

6. Using LMVT, for $x \in (-4,4)$

$$\frac{f'(x)-f'(-4)}{x+4} \geq -3$$

$$\Rightarrow f(x) \geq -2-3x \quad \ldots\ldots. \text{(i)}$$

Also $\dfrac{f'(4)-f'(-4)}{4-x} \geq -3$

$$\Rightarrow f'(x) \leq -2-3x \quad \ldots\ldots.. \text{(ii)}$$

From (i) and (ii)

$f'(x) = -2-3x \Rightarrow f(x)$ has maximum at $x = -\dfrac{2}{3}$

$$f(x) = -2x - \frac{3x^2}{2} \left[\because f(0) = 0\right]$$

$$g'(x) = f(x) = -2x - \frac{3x^2}{2}$$

$\forall x \in [0,4) g'(x) < 0 \Rightarrow g(x)$ decreasing

Also, $g''(x) = f'(x) = -2-3x < 0 \forall x \in (0,4)$

$\Rightarrow$ Graph of $g(x)$ is concave down in $(0,4)$

7. $\text{A.adj}(2\text{B}) = 16\text{I}$

$\text{A}(4\text{adjB}) = 16\text{I} \Rightarrow \text{A.adjB} = 4\text{I}$

$\Rightarrow \text{A}|\text{B}|.\text{B}^{-1} = 4\text{I}$

$\text{B.adjA} = \text{B.adj}(2\text{B}) = 4\text{B.adjB} = 4|\text{B}|\text{I}_3 = 8\text{I}_3$

$\text{A.adjB} = 4\text{I} \Rightarrow \text{A}^{-1}\text{A.adjB}$

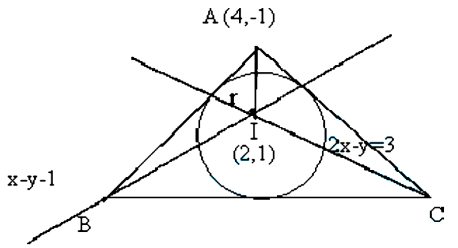
$\Rightarrow \text{adjB} = 4\text{A}^{-1}$

$$A^{-1}.adjB = A^{-1}.\left(4A^{-1}\right) = 4\left(A^{-1}\right)^2$$

$$\left(\text{A}^{-1}(\text{adjB})\right)^{-1} = \left(4\text{A}^{-1}\text{A}^{-1}\right)^{-1}$$

$$= \frac{1}{4}\text{A}^2 = \frac{1}{4}(2\text{B})^2 = \text{B}^2$$

8.



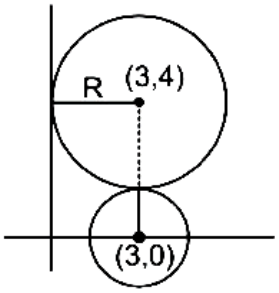
$$\text{r} = \frac{\text{AI}}{\text{Cosec}\frac{\text{A}}{2}} = \frac{2\sqrt{2}}{\sqrt{1+1/9}} = \frac{2\sqrt{2}\times 3}{\sqrt{10}} = \frac{6}{\sqrt{5}}$$

9. $xy - 3x - 4y + 12 = 0$

$(x-3)(y-4) = 0 \Rightarrow x = 3$ and $y = 4$

$S: x^2 + y^2 - 6x - 8y + c = 0$

For $R = 3$, $r = 1 \& 7$



For $R=1$, $r=3\&5$

For $R=0$, $r=4$

10. The equation of any tangent PQ to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is $\frac{x}{a}\cos\theta+\frac{y}{b}\sin\theta=1.....(1)$

The tangent cuts the ellipse $\frac{x^2}{c^2}+\frac{y^2}{d^2}=1$ at the points P and Q .

Let the tangent at P and Q intersect at the point R(h,k)

Then PQ becomes the chord of contact w.r.t.to the point R for the ellipse $\frac{x^2}{c^2}+\frac{y^2}{d^2}=1$ i.e.,

the equation of PQ is $\frac{hx}{c^2}+\frac{ky}{d^2}=1.....(2)$

The equation (1) and (2) represent the same straight line. Therefore.

$\frac{(\cos\theta)/a}{h/c^2}=\frac{(\sin\theta)/b}{k/d^2}=1\Rightarrow\cos\theta=\frac{ah}{c^2}$ and $\sin\theta=\frac{bk}{d^2}$ squaring and adding, we get

$\frac{a^2h^2}{c^4}+\frac{b^2k^2}{d^2}=1.....(3)$

Which is the locus of the point $R(h,k)$. If $R(h,k)$ is the point of intersection of two perpendicular tangents, then the locus of R should be the director circle of the ellipse

$\frac{x^2}{c^2}+\frac{y^2}{d^2}=1$, i.e., $x^2+y^2=c^2+d^2$ i.e., $\frac{x^2}{c^2+d^2}+\frac{y^2}{c^2+d^2}=1.....(4)$

The equation (3) and (4) represent the same locus. Therefore,

$\frac{a^2}{c^4}=\frac{1}{c^2+d^2}$ and $\frac{b^2}{d^4}=\frac{1}{c^2+d^2}\Rightarrow\frac{a^2+b^2}{c^2+d^2}=1$

11. $|(Z_1-Z_2)(Z_2-Z_3)|+|(Z_2-Z_3)(Z_3-Z_1)|+\|Z_3-Z_1)(Z_1-Z_3)*$

$\le|Z_1-Z_2|^2+|Z_2-Z_3|^2+|Z_3-Z_1|^2$

$\le3\left(|Z_1|^2+|Z_2|^2+|Z_3|^2\right)$

≤36

12. $\frac{S_n}{S_{2n}-S_n}=\lambda$ $t_1=S_1=1$

$\frac{S_1}{S_2-S_1}=\frac{S_2}{S_4-S_2}$ $S_2-S_1=1+d$

$\frac{1}{t_1+t_2-t_1}=\frac{1+1+d}{t_3+t_4}$

$\frac{1}{1+d}=\frac{2+d}{2+5d}$

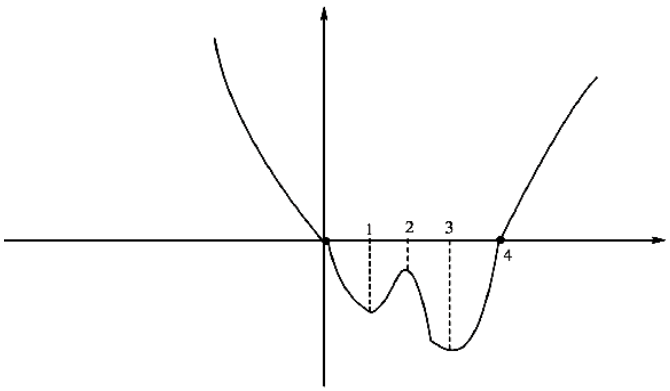
$2d=d^2\Rightarrow d=2$

13. The structure of the final order is
T_H_T_H_T_H_T_H_, and there are 4 spots to put the 2 heads in, and 4 spots to put the 5 tails in.

$$\binom{2+4-1}{2}\cdot\binom{5+4-1}{5}$$

14&15

$f'(x)=4\left(x^3-6x^2+11x-6\right)$ roots 1,2,3



16&17

$$ABC\perp DBC\Rightarrow\left(\overline{AB}\times\overline{AC}\right).\left(\overline{DB}\times\overline{DC}\right)=0\left(\bar{b}\times\bar{c}\right).\left(\bar{b}\times\bar{c}-\bar{b}\times\bar{d}-\bar{d}\times\bar{c}\right)=0$$

$$|b|^2|c|^2\frac{3}{4}-\begin{vmatrix}\bar{b}.\bar{b} & \bar{b}.\bar{d}\\ \bar{c}.\bar{b} & \bar{c}.\bar{d}\end{vmatrix}-\begin{vmatrix}\bar{b}.\bar{b} & \bar{b}.\bar{c}\\ \bar{c}.\bar{d} & \bar{c}.\bar{d}\end{vmatrix}=0\Rightarrow\frac{3|b|}{4}=\frac{9}{4}\Rightarrow|b|=3\Rightarrow|c|=3\quad|b|+|c|=6$$

18. $v = at + \dfrac{b}{t+c}$

From principle of homogeneity

$at = LT^{-1} \Rightarrow a = LT^{-2}$

$C = [T]$

$\dfrac{b}{t+c} = LT^{-1}$

$b = \left[LT^{-1}\right][T] = [L]$

19. displacement $= \dfrac{1}{2} \times 6 \times 2 - \dfrac{1}{2} \times 2 \times 2 + 2 \times 1$

$= 6 - 2 + 2 = 6m$

20. $F = \dfrac{2mv\sin\theta}{t} = \dfrac{2 \times 3 \times 100\sqrt{3}}{2 \times 0.2}$

$= 1500\sqrt{3}N$

21. Shift $x = -\dfrac{r^2 d}{R^2 - r^2}$

$r = \dfrac{R}{4}, d = R - \dfrac{R}{4} = \dfrac{3R}{4}$

$x = -\dfrac{\dfrac{R^2}{16} \times \dfrac{3R}{4}}{R^2 - \dfrac{R^2}{16}} = -\dfrac{3R^3}{64\left(\dfrac{15R^2}{16}\right)}$

$= -\dfrac{R}{20}$

22. From law of conservation of angular momentum

$mv_1 r_1 = mv_2 r_2$

$\therefore v_1 r_1 = v_2 r_2 \ldots\ldots\ldots\ldots(1)$

From law of conservation of energy, total energy of planet is constant

From (1) $v_1 = \dfrac{v_2 r_2}{r_1}$

$v_1^2 = \dfrac{v_2^2 r_2^2}{r_1^2}$

From L.C.E

$-\dfrac{GMm}{r_1} + \dfrac{1}{2}mv_1^2 = -\dfrac{GMm}{r_2} + \dfrac{1}{2}mv_2^2$

From this we get $v_2 = \sqrt{\dfrac{2GMr_1}{r_2(r_1 + r_2)}}$

Angular momentum $L = mv_2 r_2 = m\sqrt{\dfrac{2GM\ r_1 r_2}{r_1 + r_2}}$

23. $mV = (m+m)V$

$\therefore\ V = \dfrac{\vartheta}{2}$

From conservation of energy

$\frac{1}{2}m\vartheta^2 = \frac{1}{2}(2m)v^2 + \frac{1}{2}kx^2$

$\frac{1}{2}m\vartheta^2 = m\frac{\vartheta^2}{4} + \frac{1}{2}kx^2$ .....(1)

$\therefore x = \vartheta\sqrt{\frac{m}{2k}}$

From (1) $\frac{1}{2}kx^2 = \frac{m\vartheta^2}{4}$

24. The reaction of efflux of liquud $= \frac{m}{t}\vartheta$

$(\mathrm{Ah}\,\rho)\alpha = \rho a\vartheta^2 = \rho a(2gh)$

$\therefore \alpha = \frac{2ag}{A}$ .....(1) α is acceleration

Time taken for the tank to be empty

$t = \frac{A}{a}\sqrt{\frac{2h}{g}}$

$v = \alpha t = \sqrt{2gh}$ .....(2)

25. Frequency of fork B

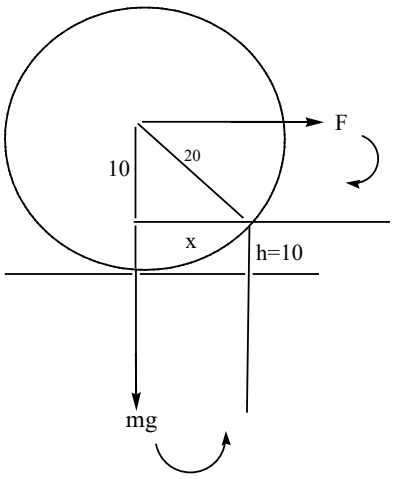
$n_B = n_A \pm 8 = 258 \pm 8$

$= 266\,hz\,(\mathrm{or})\,250\,hz$

Here known fork is filed and beats decreased

$\therefore n_B = 266\,hz$

26.



$x = \sqrt{400-100} = 10\sqrt{3}\,cm$

c.w torque =A.c.w torque

$F \times 10 = mg \times 10\sqrt{3}$

$F = 10 \times 10\sqrt{3} = 10\sqrt{3}\ kgwt = 10 \times 1.732 = 17.32 kgwt$

$17 kg\ wt$

27. Dispersive power $w = \frac{\mu_F - \mu_C}{\mu_D - 1}$

$= \frac{1.535 - 1.527}{0.53} = \frac{0.008}{0.53}$

$= 0.015$

28. $C_1 = \frac{Q}{V} = Cot\ 30^0$

$C_2 = \frac{Q}{V} = \tan 60^0$

$\frac{C_1}{C_2} = \frac{Cot 30^0}{\tan 60^0} = 1$

29. $V_{rms} = \sqrt{\frac{4+9+16+25+36}{5}} = \sqrt{18} = 4.24$

30. $K = \frac{\Delta P}{\left(\frac{\Delta V}{V}\right)}$ $\Delta P = P_2 - P_1 = (1.165 - 1.01)$ x 10^5 $= 0.155$ x 10^5

$= \frac{0.155 \times 10^5}{10^{-1}}$ $\frac{\Delta V}{V} = \frac{10}{100} = 10^{-1}$

$= 1.55 \times 10^5$

$n = 1.55$

31. $\frac{d\rho}{dT} = \alpha\ \rho = -0.0005 \times 3.5 \times 10^{-5}$

$= -0.00175 \times 10^{-5}$

32. $n = -\alpha T = -\left(-0.0005\right) \times 300$

$= 0.15$

33. After the steady state reached, current through R, immediately after S is opened is zero

34. Through R_2

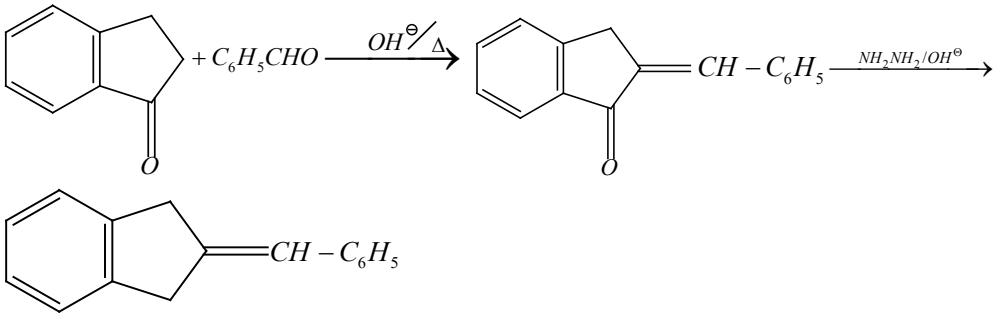
$i_2 = \frac{50}{22} - \frac{15}{11} = \frac{50-30}{22}$

$= \frac{20}{22} = \frac{10}{11}$

$= 0.909$

CHEMISTRY

35.



First Step is crossed Aldol reaction and second step is Wolff kishner reduction.

36.

$$H_2C_2O_4 \xrightarrow{\Delta} \underset{(A)}{CO} + \underset{(B)}{CO_2} + H_2O$$

$$CO + Cl_2 \longrightarrow \underset{(D)}{COCl_2} \xrightarrow{NH_3} NH_2CONH_2$$

37. $X_0 - \frac{X_0}{2}$ (Zero Order)

Time needed $\frac{X_0}{2k}$

$\frac{X_0}{2}$ *to* $\frac{X_0}{16}$ [First order]

Time needed = 3 half life periods

$= 3\frac{\ell n2}{k}$

Total time = $\frac{X_0}{2k} + \frac{3\ell n2}{k}$

38. Since $E.M.F_{cell}$ to be zero

$$E^{O}_{Ag^+/Ag} + 0.059 \log \frac{k_{sp\,AgCl}}{[Cl^-]} = E^{0}_{Ag^+/Ag} + 0.059 \log \frac{Ksp\ AgBr}{[Br^-]}$$

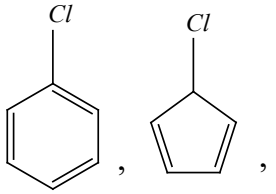
$$\frac{[Br^-]}{[Cl^-]} = \frac{K_{SPAgBr}}{K_{SPAgCl}} = 5\times10^{-3} = 1:200$$

39. Those carbohydrates in which configuration at C_3, C_4, C_5.... Carbon atoms are same give same Osazone

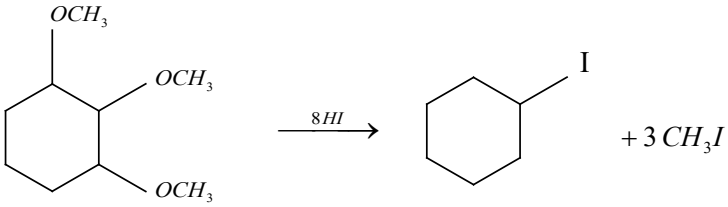
40. Conceptual

41. Conceptual

42.

Except:  , , $H_2C = CH - Cl$

43.



44. Work done at constant
Pressure = nRdT = 25
$C_v = 3R,\ \ C_P = 4R$ Since average degree of freedom = 6
Total heat absorbed = nC_pdT
$= n\ 4RdT$
$= 4 \times nRdT$
$4x^2 = 100\ J$
$x = 5$

45. $\dfrac{\text{Solubility in presence of acid}}{\text{Solubility inPresence of water}} = \sqrt{1 + \dfrac{\left[H^+\right]}{Ka}}$

$\dfrac{61.8}{8.35} = \sqrt{1 + \dfrac{10^{-3}}{ka}}$

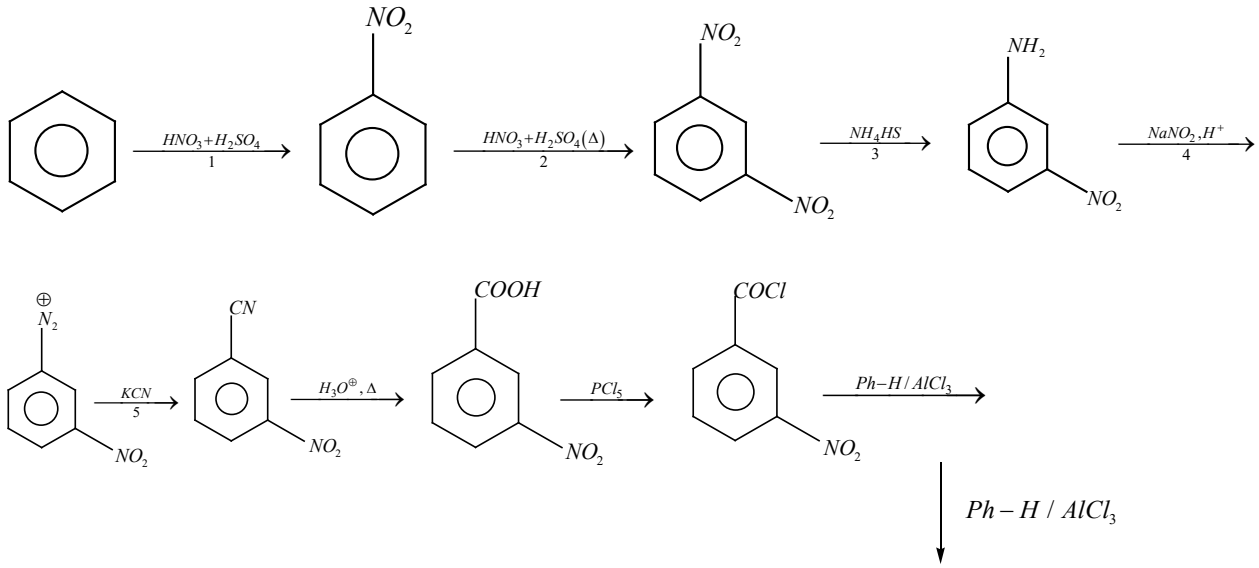
$Ka = 1.85 \times 10^{-5}$

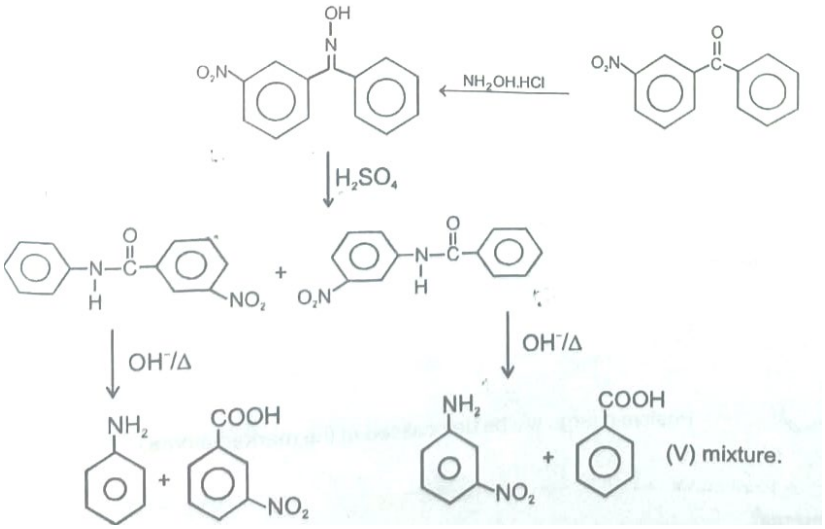
x value is 5

46. $-\dfrac{\Delta G^O}{2.303\ RT} = \log k \qquad k = \dfrac{k_1}{k_2} = 10^3\ \ \log 10^3 = 3$

47. Acidified $K_2Cr_2O_7, CuSO_4, H_2O_2, Cl_{2,}O_3, FeCl_3, HNO_3$ can oxidize Iodide to Iodine.

48.





50. $\frac{1}{m} = \frac{d}{m} - \frac{m.wt}{1000} \qquad = \frac{1.051}{16.8/11.2} - \frac{34}{1000} = 1.5$

51. Wt of H_2O = total weight – weight of O_2
= 5255 – 120 = 5135